



BRAC UNIVERSITY

CSE461: Introduction to Robotics

Lab Project Report

Title: Autonomous Garbage-Collecting Robot with Claw

by

[Group_5]

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Abstract

The “**Autonomous Garbage-Collecting Robot with Claw**” is a low-cost robotic system that automates the collection of small to medium particles of garbage from indoor and semi-indoor environments. The robot is built using an Arduino Uno microcontroller, L298N Motor Driver, DC gear motors and the HC-SR04 Ultrasonic Sensor, which is used to detect obstacles and assist the robot to navigate without collision. The ESP32-CAM Wi-Fi and Bluetooth camera module is used to detect garbage objects, and the robot uses a servo motor to open and close its mechanical claw to retrieve the object from the garbage and deposit it in a garbage container within the robot. The weight of the collected waste is measured with a load cell attached to an amplifier module (HX711), and a buzzer is used to give an alarm when the waste container is full. An LM2596S DC-DC step down buck converter is also included, to ensure stable and regulated power to the needed components. For sustainability and real-world impact, the system can be used to improve the cleanliness of shared spaces, cut down on manual litter collection and unnecessary human contact with unhygienic waste, and could be upgraded in the future with fewer material costs for the reusability of its modular components. For larger-scale deployment, however, the following parameters still warrant consideration: energy consumption, use of the camera, worker impact and safe usage around human beings.

Keywords

[Autonomous robot, waste collection, Arduino Uno, ESP32-CAM, robotic claw]

1. Introduction

Garbage collection in commonly used areas like campus, classroom, corridor etc. is still largely done manually. It is repetitive, time consuming and can be unsanitary for cleaning staff. The motivation for this project was the potential to use the simple autonomous robot to help with collecting garbage and to minimize repetitive manual work for small scale garbage collection.

The primary goal of the project is to create an Autonomous Garbage-Collecting Robot with Claw that will navigate around an environment, avoid large obstacles, identify small objects on the ground, retrieve them with a mechanical claw and deposit them in a garbage container on board. The robot also keeps track of the waste that it collects and gives an alarm when the waste reaches a set amount.

The project is primarily intended for flat surfaces indoors and semi-indoors where it can capture lightweight wastes like bottle caps, paper and plastic wrappers. It is relevant to robotics and computer interfacing as it can integrate sensors, microcontroller, motors, servo-controlled

mechanisms, wireless camera and regulated power management. The ESP32-CAM module not only brings the visual, but also the wireless capabilities to the system, while the LM2596S module supplies the correct regulated voltage to the rest of the components. The Arduino Uno module works as the main processor on this device, picking up sensor readings and driving the actuators and the ESP32-CAM module brings both visual and wireless functionality, supplying the right regulated voltage to the rest of the system with the help of the LM2596S module.

2. Related Work/ Inspiration

This project is motivated by autonomous cleaning and waste-collection robots, which show that mobile robotic systems can be used for partially automating repetitive cleaning jobs. Current robots for cleaning use a combination of sensors, microcontrollers, motors, and obstacle avoidance systems to move around an environment where humans are present to a limited extent. More advanced waste-collection systems may also use cameras and image-processing techniques to identify objects before performing collection. These concepts inspired us to design a small scale autonomous robot that would be able to navigate in a controlled environment, locate waste, pick up the waste mechanically and store it in a container mounted on the robot.

The uniqueness of our project is the way it incorporates a number of functions into a fairly inexpensive prototype. The ESP32-CAM is used to detect the garbage, and the HC-SR04 ultrasonic sensor is used to detect obstacles during navigation. The Arduino Uno, L298N motor driver and DC motors control the movement of the robot and the servo controlled claw is used to physically collect the garbage detected, not just cleaning the surface of the floor. A load cell and HX711 module is also used to track how much waste is collected, and alerts using a buzzer once the waste container is full. So, it integrates autonomous navigation, garbage detection via camera, mechanical garbage collection and container capacity monitoring into one robotic platform.

3. Technical Approach

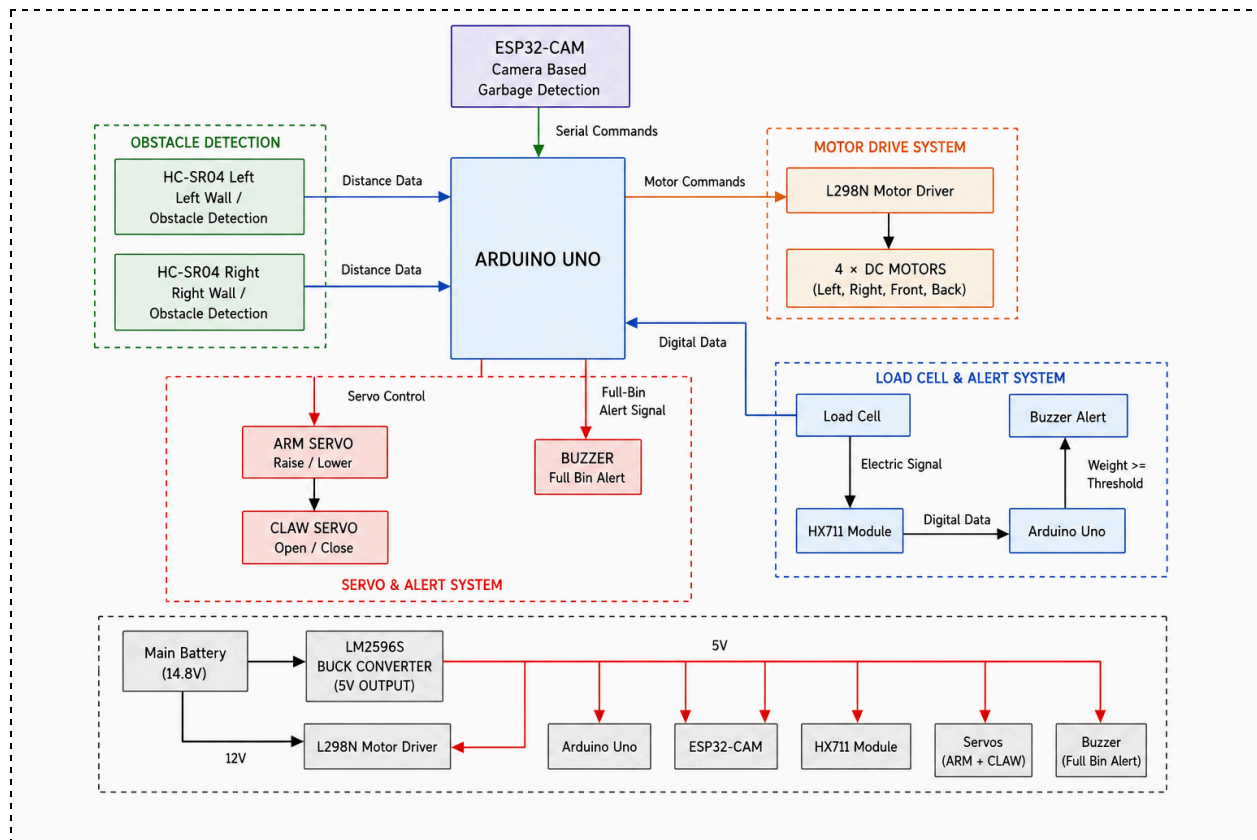
System Architecture:

The robot is designed using a modular approach with the Arduino Uno microcontroller as the central processing unit. The autonomous serpentine navigation is achieved by using two HC-SR04 ultrasonic sensors which returns wall and obstacle distance information, while the ESP32-CAM is used to detect garbage using the camera and send detection command data to the Arduino via serial communication. These inputs are used by the Arduino to drive the four DC motors through the L298N motor driver and also control the two SG90 servo motors that control the arm and claw mechanism. The load cell is used to detect the weight of garbage collected

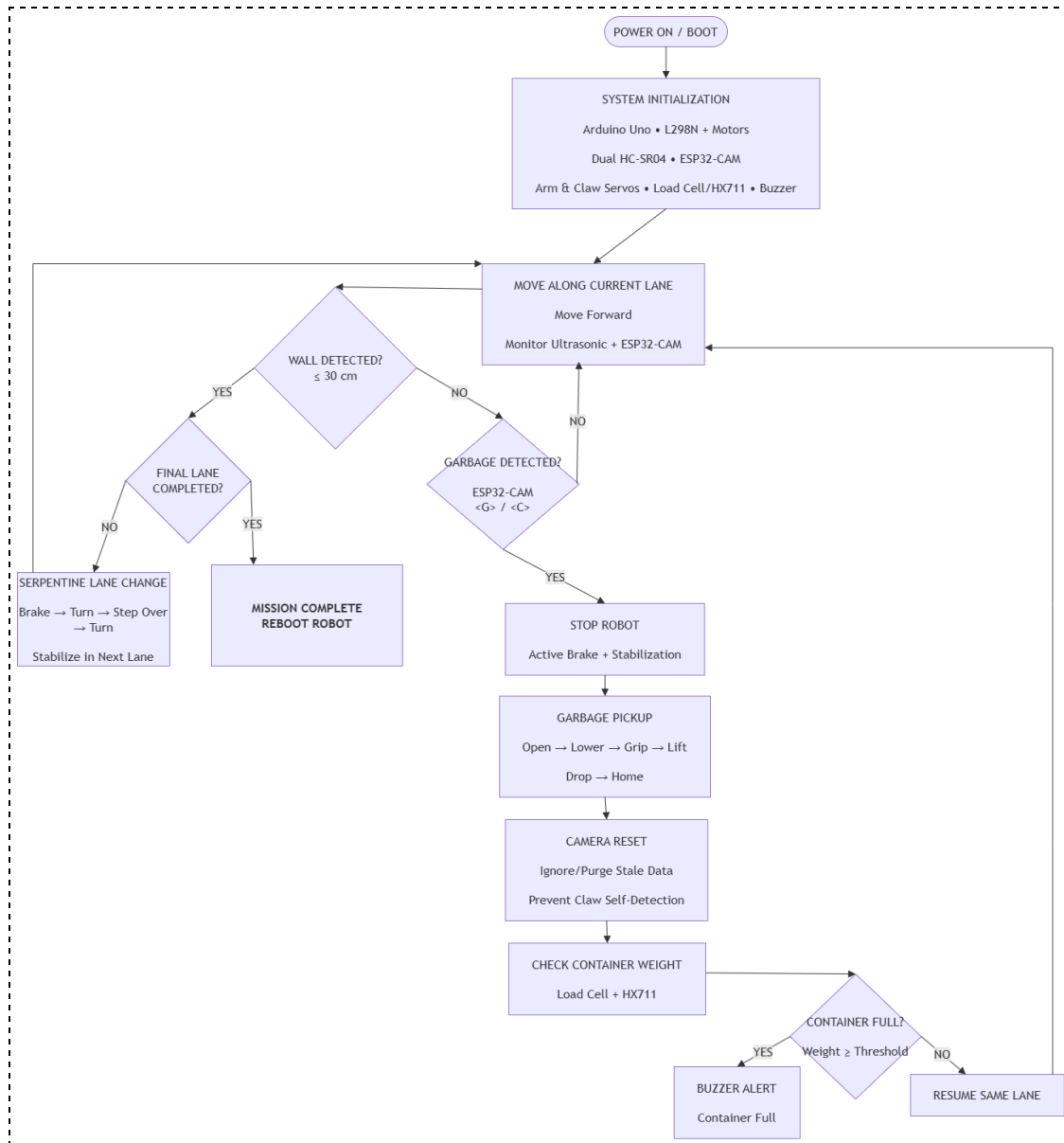
which is fed into the HX711 amplifier module, and when the garbage container reaches the pre-set weight, the buzzer will be activated by the Arduino.

Several parts are separated in the power system to operate safely and avoid the initial issue with the ESP. The main power provides electricity to the Arduino Uno and the L298N motor driver needed to drive the robot. The arm and claw motors need more stable power during operation, so a separate regulated line from the LM2596S buck converter is used to power them. To reduce the instability of power during operation, the ESP32-CAM is designed to be powered separately from the power bank. While various types of power sources are employed, they all have a common ground reference for correct communication and control.

BLOCK DIAGRAM:



Work Flow Diagram:



Components Used:

Hardware Tools:

- **Arduino Uno R3:** Acts as the main controller of the robot. It reads the sensor inputs and manipulates the motor driver, servo motors, load-cell system and buzzer based on the programmed logic.

- HC-SR04 Ultrasonic Sensor: It is used to detect obstacle and avoid it. Measures distance between the robot and objects in front of it continuously to allow the robot to change its movement if it detects an object.
- ESP32-CAM Wi-Fi + Bluetooth Camera Module: For camera based garbage/object detection. It offers visual data to help the robot identify an object to retrieve, as well as built-in Wi-Fi and Bluetooth.
- ESP32-CAM-MB Micro USB Module: This module is used to upload codes and interface with ESP32-CAM through micro USB, which makes the coding and development process easier.
- Load Cell Weighing Sensor: Mounted under the garbage can to detect the weight of the garbage collected and when the garbage can will exceed the set capacity.
- HX711 Load Cell Amplifier Module: Connects the load cell to the Arduino, amplifies and digitizes the small electrical signal from the load cell to be used by the controller.
- LM2596S DC-DC Step Down Buck Converter: Steps down and regulates the voltage supplied from the power source to a proper level for the electronics, so that electronic components have a more stable power supply.
- SG90 Servo Motors (180°) ×2: To drive mechanical claw mechanism. The opening and closing of the gripper and the raising and lowering of the claw arm are controlled by one servo and the other respectively.
- L298N Motor Driver Module: It is used to control the DC motors with the Arduino Uno. It lets the Arduino to control the direction and movement of the robot.
- The 4WD Chassis and Wheels : It builds the body of the robot and the base on which the controller, motors, sensors, camera, claw, garbage container, batteries, and more will be attached.
- Buzzer Module: Gives an audible warning when the load-cell reading is at the set capacity of the garbage container.
- 9V Batteries with Holders ×4: Provide electrical power to the robot and its electronic components.
- Breadboard: To connect the various electronic modules in a temporary circuit and to connect the components.
- Jumper Wires: These wires are used for making the electrical connections between the Arduino Uno, Sensors, Motor Driver, Load-cell module, buck converter, and other electronic components.
- Mechanical claw: Collects the garbage and is mounted on the robot. Plastic Bucket/Container: Stores the garbage collected by the mechanical claw and is mounted on the robot.
- Miscellaneous Components: These include screws, foam, zip-ties and other mounting parts that attach the electronic and mechanical components to the robot chassis.

Software Tools:

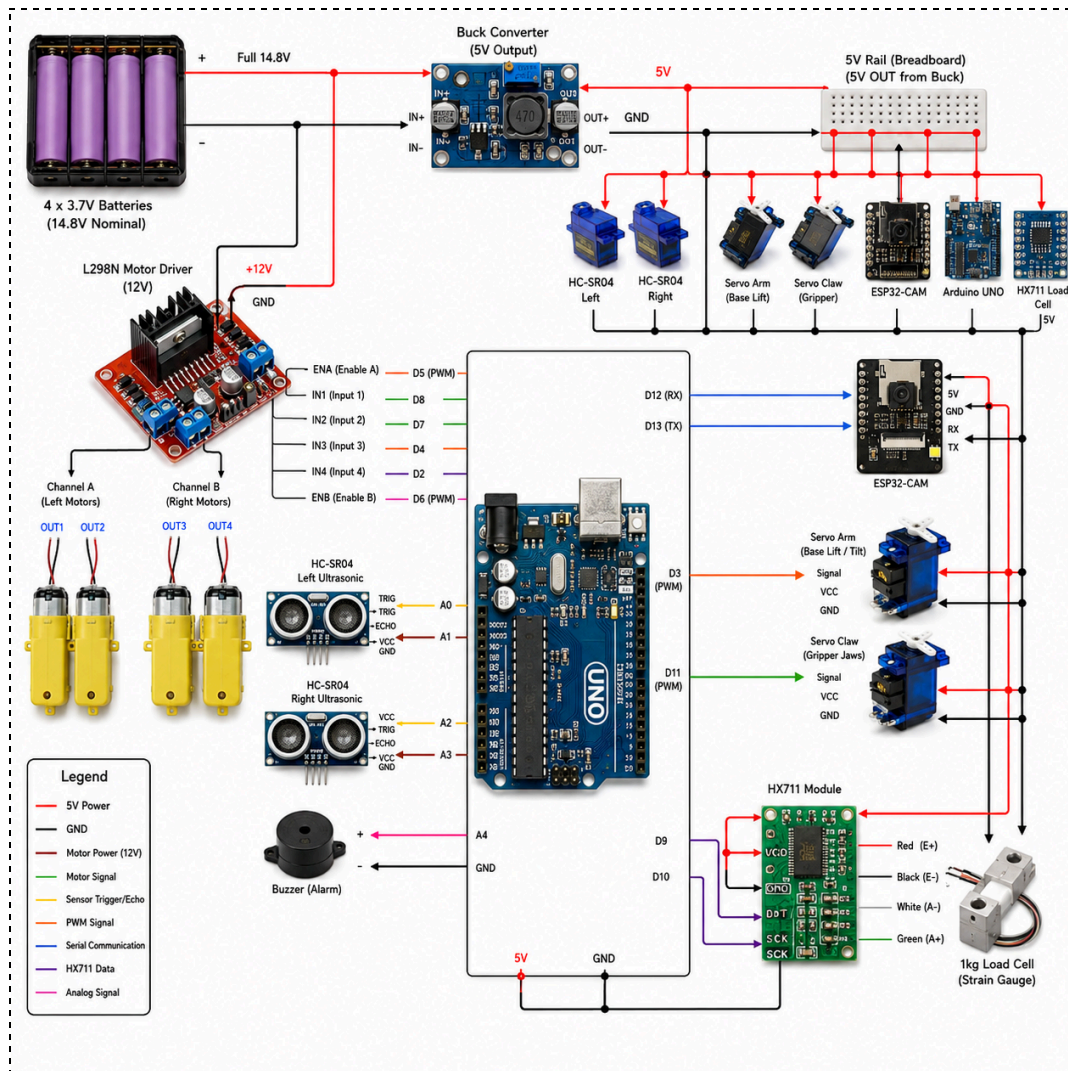
- **Arduino IDE:** Arduino programming environment for writing, compiling and uploading Arduino Uno and compatible ESP32-based components. The programmed logic controls the robot's navigation, obstacle avoidance, garbage collection, weight monitoring and other functions.

Cost Breakdown (Approx.)

Component	Unit Cost (BDT)	Quantity	Total (BDT)
Arduino Uno R3	990	1	990
HC-SR04 Ultrasonic Sensor	100	2	200
ESP-CAM WiFi+Bluetooth Camera Module	950	1	950
ESP32-CAM-MB Micro USB Module	220	1	220
Loadcell Weighing Sensor	280	1	280
LM25965 Buck Converter	380	1	380
Load Cell (HX711 Module)	195	1	195
SG90 Servo Motor (180°)	135	2	270

L298N Motor Driver Module	220	1	220
4WD Chassis Wheels	1,116	1	1,116
DC Motors	95	4	380
Jumper Wires (pack)	120	3	360
Breadboard	160	1	160
9V Battery + Holder	90	4	360
Buzzer Module	20	1	20
Plastic Bucket / Container (for garbage bin)	20	1	20
Miscellaneous (screws, foam, zip ties)	100	1	100
Total Estimated Cost			6,221

- **Circuit/Schematic:**

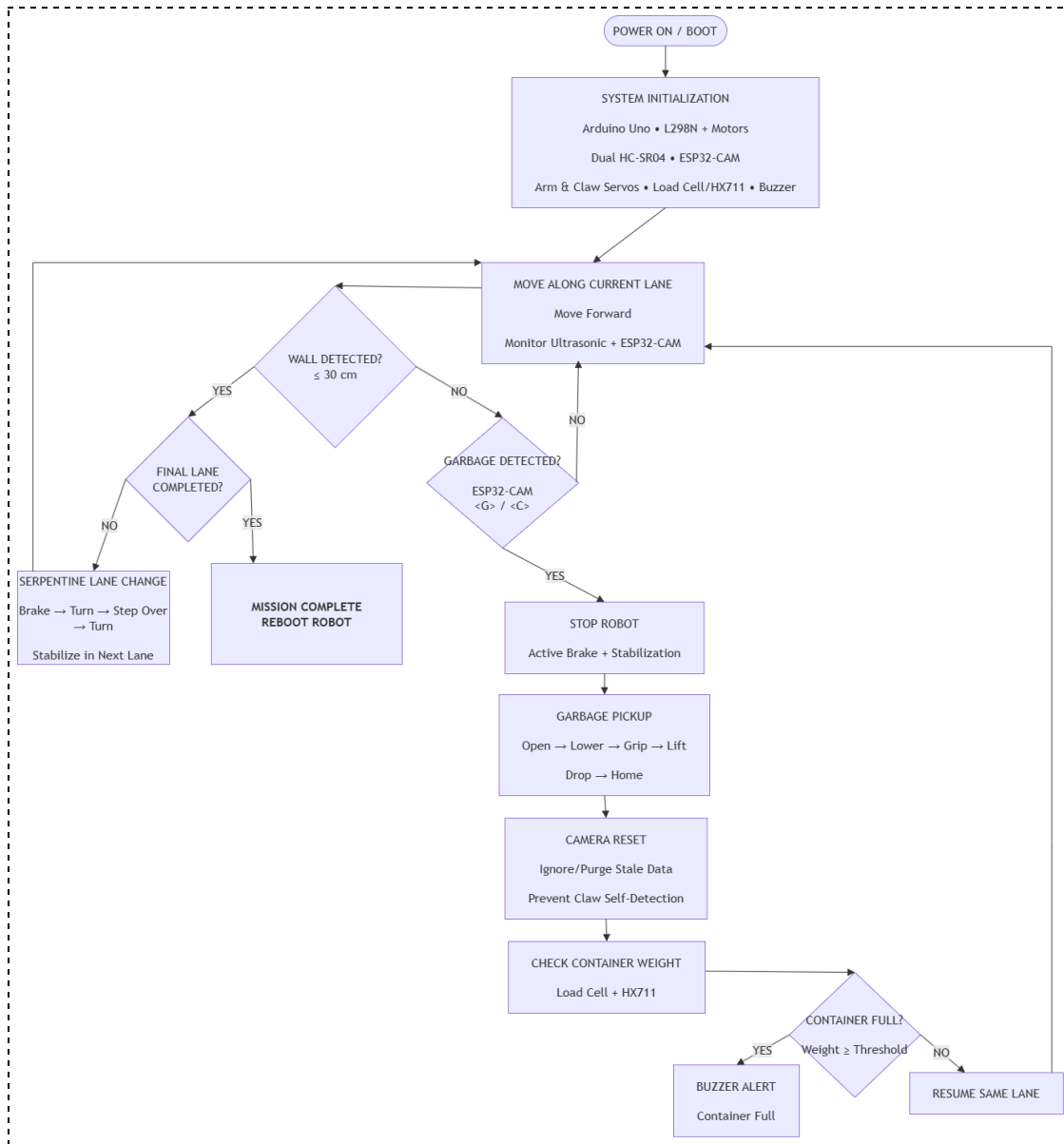


Functionality:

The robot works by a series of autonomous navigation, camera-based garbage detection, mechanical garbage collection, and container capacity monitoring. The distance data from the two HC-SR04 ultrasonic sensors is fed into the Arduino Uno which will be the primary controller and detection commands will be sent to the ESP32-CAM. The Arduino uses these inputs to control the L298N motor driver, four DC motors, two servo motors, load-cell system and buzzer. The ESP32-CAM is connected to the Arduino using a serial communication and it generates the garbage-detection signal that triggers the garbage-picking sequence. The previous report already specifies the ESP32-CAM as the

garbage-detection subsystem and the load cell/HX711 as the container-monitoring subsystem.

WorkFlow Diagram



- **System Initialization:** After power-on, the Arduino Uno initializes the motor driver, dual ultrasonic sensors, ESP32-CAM communication, servo motors, load cell with HX711, and buzzer. Until a valid target is detected, the arm and claw will stay at home position.

- Lane Navigation: The robot follows the lane with the use of four DC gear motors powered by the L298N motor driver. Distance from walls and obstacles are constantly measured by the two HC-SR04 sensors.
- Wall Detection and Lane Change: If a wall is detected within the specified stopping distance, the robot slows down and performs a serpentine lane change manoeuvre: first turn, step lateral, second turn and stabilisation. It then goes along the next street in the opposite direction.
- Camera-based Garbage Detection: ESP32-CAM is detecting garbage in front of the robot during robot movement. If a target is detected and located correctly, the ESP32-CAM will transmit a serial data over serial communication to Arduino UNO to indicate that a target is detected. The commands used in the designed protocol include a <C> for centred target and a <G> for the pickup position.
- Garbage Collection: As soon as the robot receives the necessary detection command, it comes to a halt and stabilizes. The claw opens, the arm moves towards the garbage, the claw closes and picks up the object, the arm carries the object up to the rear garbage container and drops off, and the arm returns to its home position. The camera input is disabled when the arm moves so as not to make a false detection due to the movement of the claw.
- Post-Pickup Camera Reset: When an object is dropped, the old camera packets are cleared, and then the camera is reinabled for detection. This helps to avoid a previous detection, floor reflections or movement of the claw from triggering a new pick-up sequence as soon as it happens. Container
- Weight Monitoring: Under the garbage container is connected to the load cell, which delivers the load to the HX711 module, and then to the Arduino. The measured weight is constantly monitored against the set capacity limit. Full-Bin Alert: If the container weight nears the set weight, the buzzer sounds which alerts the public that the garbage is full.
- Resume Navigation: If collection can proceed then the robot will return to the same lane from where garbage collection was interrupted and continue to search.
- Mission Completion: The robot will complete the process of searching the serpentine lanes and picking up garbage over the designated search area, until the last lane is cleared.

Challenges:

- **ESP32-CAM Hardware failure:** The most prominent issue was the frequent failure of the ESP32-CAM subsystem in the initial design, that was used for garbage detection using a camera. The first ESP32-CAM got damaged due to using the same power line for the motor system and ESP32-CAM. The DC motors caused electrical noise and a lot of fluctuations in current when moving, which caused the power supply to become unstable for the ESP32-CAM and later destroyed it.

The power system was being separated for the second ESP32-CAM to overcome the same issue. The battery supply was used to power the servo motors and the power bank was used to power the ESP32-CAM. The second ESP32-CAM was also damaged at the end of the integration without being connected to a power source. There was no definite explanation for this second failure.

The second failure happened on the last day of the development period; there was not enough time to get another ESP32-CAM, build-up the vision system, and set-up the communication between claw and camera. Hence, the final prototype was built using the original design – the ESP32-CAM, while a simple IR sensor was utilized as an emergency fallback, for basic garbage/object detection.

- **Ultrasonic Sensing:** When the robot was coming close to smooth or angled walls, the HC-SR04 sensors would sometimes send false readings, due to the ultrasonic waves echoing off of the wall and not reaching the receiver. The stopping distance and threshold were, therefore, adjusted by testing and two front ultrasonic sensors were added to enhance frontal and corner coverage.
- **Servo Jitter and Mechanical Calibration:** The arm and claw servos exhibited jitter, unwanted motion and excessive travel initially. The unnecessary movement and power consumption was minimized by physically calibrating the servo angles, slowing and restricting the servo movements, and detaching the servos when not in use. During arm motion, also camera input was neglected to avoid false detection of the claw.
- **Power-Supply Instability:** The servo mechanism was found to be more reliable when powered by the Arduino through USB than battery power. With the voltage drops and unstable movement due to the higher current demand on the servos. To ensure a more stable regulated power supply, an LM2596S buck converter was selected, and attention was paid to the power distribution and common-ground connections of the Arduino and servos with the ESP32-CAM.

4. Sustainability & Impact

Sustainability: This autonomous garbage-collecting robot can help achieve environmental sustainability by assisting in the regular garbage collection service from shared areas like offices, classrooms, corridors, and campuses for small to medium garbage. The system can help to reduce the accumulation of waste, which can lead to cleaner environments and improved waste management. Most of the hardware components such as Arduino Uno, ESP32-CAM, ultrasonic sensor, motor driver, servo motors, DC motors, load cell, and chassis are reusable and can be replaced individually, minimizing the need to replace the entire system if one component needs to be replaced. The LM2596S DC-DC step-down buck converter also supplies regulated voltages to the necessary components, ensuring stable and efficient power distribution across the system.

Impact: The robot can save repetitive manual labour in picking up small pieces of litter and minimise direct contact between humans and potentially unhygienic waste. In the future, therefore, cleaning workers might spend less time picking up small waste items and more time on human activities like deep cleaning, dealing with larger waste, and facility maintenance. A robotic claw for waste collection could also minimize worker's exposure to dirt, bacteria, sharp objects and other potential waste contaminants. In addition, the regular collection of litter can help maintain cleanliness and the aesthetic appeal of commonly used spaces, like campuses, workplaces, corridors, walkways, etc which are more pleasant for those who use them.

Future Work: To improve the system further, the accuracy of the garbage detection using the ESP32-CAM can be enhanced, allowing the robot to accurately identify waste from other small objects or distinguish different kinds of garbages and keep them separate. For example, biodegradable, non-biodegradable, wet etc. Further image processing or machine learning could be used for better object classification, adding distance sensors and enhanced path planning algorithms would increase efficient navigation and obstacle detection. The mechanical claw may also be enhanced to deal with the various shapes and sizes of objects. Future versions of the robot may incorporate rechargeable batteries, battery level monitoring, automatic battery charging, automatic wireless status monitoring, and automatic return mechanism that will enable the robot to go to a designated location if the garbage container is full. For more secure use in the real world, an emergency stop could also be included.

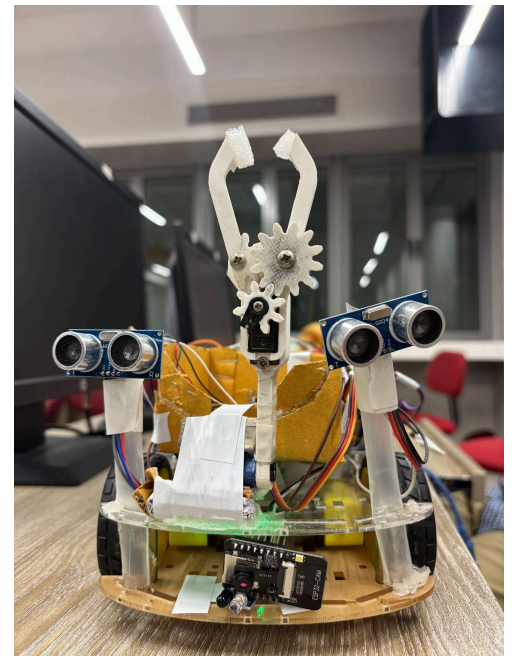
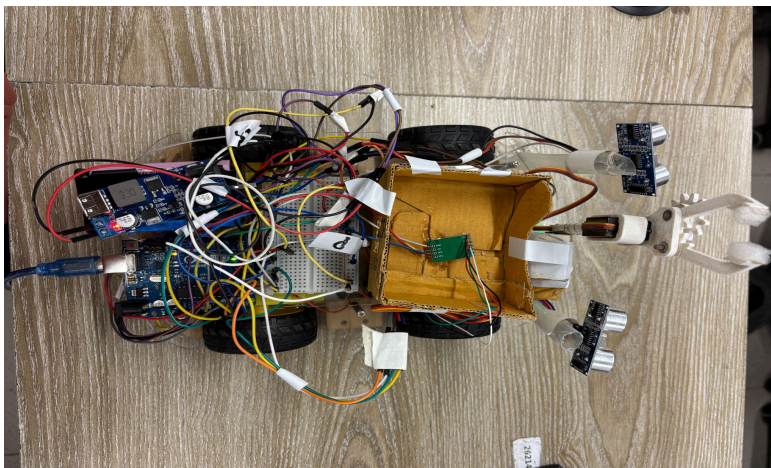
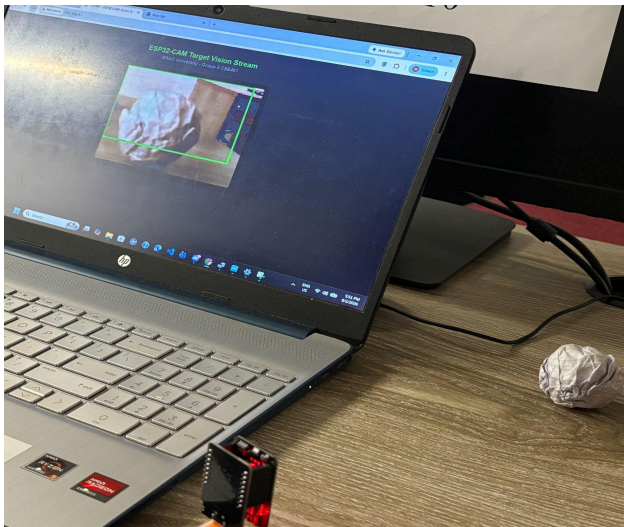
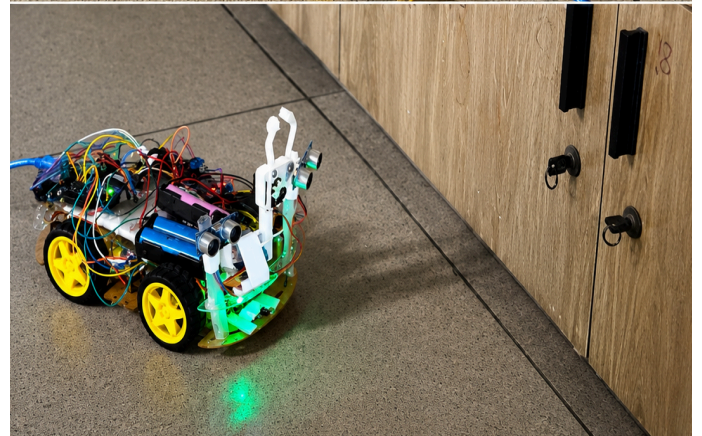
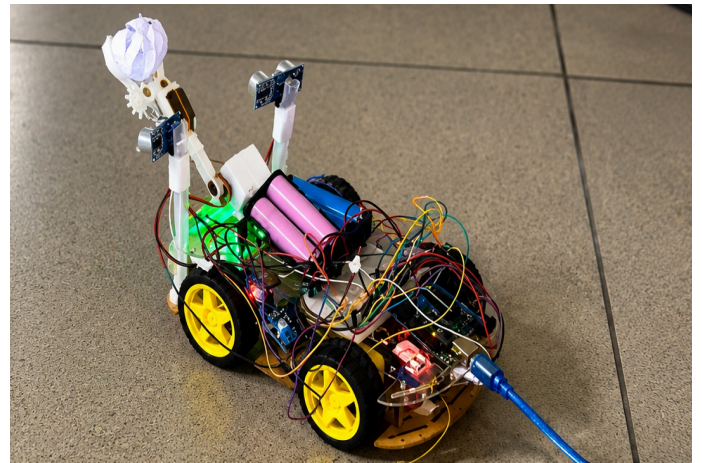
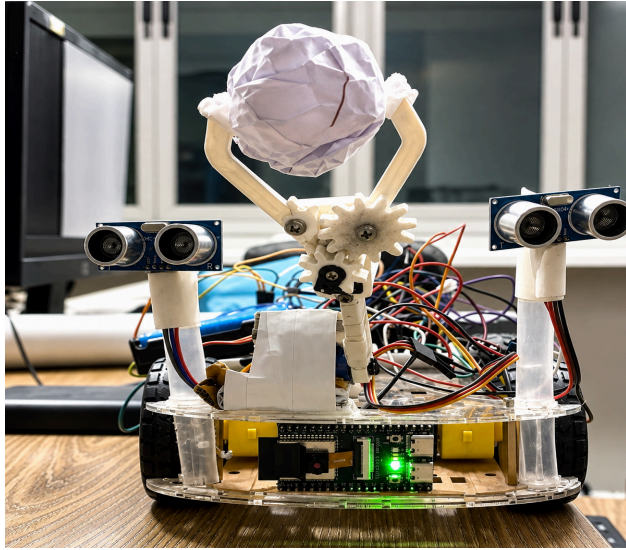
Limitations: The prototype currently under development has some hardware and software restrictions. The ESP32-CAM is used for garbage detection, so the results might change under different lighting, camera angle, visibility and waste object appearances. It is basically used to detect obstacles in front of the robot, hence it has lesser awareness of obstacles coming from other directions. The mechanical claw is optimized for small, light garbage, and may not reliably pick up heavy, irregular shaped, slippery and sharp objects. The robot also cannot be used outdoors or on stairs or rough ground surfaces or in very crowded conditions. Furthermore, continuous operating time is limited by battery capacity and safety issues around persons are

caused by the moving robot and claw. If the robot is intended for use in shared or public spaces, privacy and responsible use of camera data needs to be taken into account as well.

5. Results & Discussion

The prototype was tested on the various functions: navigation, garbage detection, claw operation, and power stability. The key results are summarised below.

Function	Outcome
Robot Movement	DC motors and L298N were able to control robot movement and turning
Ultrasonic Navigation	Dual HC-SR04 sensors were used for wall detection, but sometimes had inconsistent readings on angled surfaces.
Arm and Claw	To minimize jitter and unnecessary movement, servo angles were calibrated
ESP-32 Cam	Two ESP32-CAM modules were damaged, and the camera detection was not retained.
Power System	A more stable regulated supply to the servo system was provided successfully
Bin Monitoring	Load cell and HX711 were used to monitor bin weight and set up the buzzer.



Overall, it was shown that the prototype could move independently, detect objects, mechanically pick them up, and monitor the bin. But there were a few points that need further improvement such as ESP32-CAM failure, limitation of ultrasonic sensor and power instability.

6. Conclusion

The Autonomous Garbage-Collecting Robot with Claw showcases how robotics can be applied to solve a real-world problem by combining autonomous navigation, object detection, mechanical waste collection and garbage capacity monitoring. These fulfill the purposes of detecting garbage without manual process in lower cost also. The main system is controlled by Arduino Uno with ultrasonic sensor and IR sensor as input and it includes DC motors. A mechanical claw is controlled by servo motors and a load cell with HX711 measures the weight onboard garbage container. The addition of the ESP32-CAM gives camera and wireless functionality to the board and regulated power distribution is enhanced with the LM2596S buck converter.

So, the project illustrates a fairly inexpensive way to combine robotic components to help with an iterative real world task. The system offers potential benefits in the aspect of cleaner shared spaces, saves man power and reduces direct contact with unhygienic waste. Although the current prototype has issues to recognize object, grip, sensor accuracy, power capacity and safety. These constraints offer definite future scope for improvement to make it more reliable and intelligent autonomous waste-management system.

7. Contribution

Name	ID	Role(s) / Responsibilities	Specific Contributions
Alena Halder Tamo	22301046	Hardware Design, Documentation	• Assisted with circuit connections and component setup • Worked on sensors and power-system integration • Participated in hardware testing and debugging

Name	ID	Role(s) / Responsibilities	Specific Contributions
			• Prepared report, diagrams, and documentation
Tahmida Tanjim Talukder	22101730	System Integration, Circuit Integration	• Integrated overall system design and workflow • Worked on sensor–controller integration • Participated in hardware testing and debugging • Prepared report, diagrams, and documentation
Mofftasim Hossain Sayem	22299109	Software Development, Navigation	• Contributed significantly to Arduino programming and software integration • Worked on navigation, motor-control, sensor-processing, and system logic • Assisted with hardware integration and debugging • Contributed to testing and technical documentation
Tahsin Raihan Robbani	22201344	Mechanical Design, Claw System	• Worked on mechanical claw and servo setup • Calibrated arm and gripper movement • Participated in calibration, troubleshooting, and documentation

Name	ID	Role(s) / Responsibilities	Specific Contributions
Mayen Shahriar Kabir	23101107	ESP32-CAM / Sensor Integration, Hardware, Testing	Assisted with the overall assembly of the robot and circuit construction • Worked on Arduino, L298N motor driver, DC motor, ultrasonic sensor, servo motor, ESP 32 cam and power-line connections. • Assisted with debugging hardware, prototype testing and final system integration • Assisted in the development of control logic and report writing .

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